INFO4555/5556 Business Intelligence Systems
Fall 2025 Class Project Description

Project: Cornell Data Warehouse & Data Lakehouse Access & Usage Reporting

Project Sponsors:

Ben Maddox – Chief Information Officer

Bobby Edamala – Chief Information Security Officer

Lisa Clark – Office of the University Registrar, Deputy University Registrar

Curtis Josey – University Audit Office, Associate Director of Analytics & Strategy

Norah Selig – Alumni Affairs and Development, Director of Data Strategy & Enterprise Systems

Ryan Brown – Department of Financial Services, Associate Controller

Key Dates:

October 9th Requirements refinement session with students

December 2nd & 4th Team presentations to sponsors

Background:

Cornell University's Data Warehousing environment is a complicated system comprising of many separate databases and more than one type of database (Oracle, Postgres, Redshift). The architecture allows for great flexibility and scalability, however, there is no single view into who has access to which data and who is accessing what data. There are three groups who need access to this information about data access and usage: IT Security office, University Audit Office, and the Data Stewards. Currently these groups send requests for this data and it is provided to them via custom generated spreadsheets. The process to pull this data, from the many different databases, takes time and the data is only pulled a few times a year. There is currently an effort underway to pull the needed access and audit data, from each of the underlying databases, into the Data Lakehouse. Once this data is centralized, there will be data modeling needed to properly merge and represent the data. Finally, there will be a need for dashboard(s) for the IT Security office, Audit Office, and Data Stewards to provide convenient access to the Security and Audit data.

Goals of this Student Team project:

Using a sample of the security mapping and audit data, come up with Proof of Concept BI systems to inform a later IT project. Student teams will create data models, working ETL code, and sample dashboards.

Student teams will initially use this high-level requirement document to get started. On October 9th, our project sponsors will come to class to answer questions the student teams have come up with regarding the requirements and data. The session will be used to refine the requirements the student work from.

At the end of the semester, student teams will present their working Proof of Concept BI systems to the sponsors. As a deliverable, the student teams will submit documentation including: data models, source-to-target mappings, data dictionary, and dashboard documentation.

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This project is an opportunity for students to exercise the skills they are learning in class, using real data and real BI requirements. For Cornell, this is an opportunity to explore options for deploying a long-term BI solution.

Known requirements:

A dashboard, or dashboards, to be used by designated staff from the IT Security Office, University Audit Office, and Data Stewards containing data access and usage information encompassing the entire Cornell Central Data Warehouse and Data Lakehouse environment.

Attributes of interest include:

- Database account name (may be a netid or generic application account)
- Account owner
- Account owner college(or unit)
- Account owner department
- Account owner employee status (A=active, I=inactive, R=retired)
- Database account located
- Database roles (actual DB role name) the account has access to
- User friendly role names
- User Friendly data mart name (hierarchy - data mart/role/table)
- The date/time account last used each assigned role
- Number of times (queries) the account ran using each assigned role (in last 3 months)
- Lower grain details (drill down from role access info):
    - Similar to user role access, but user individual table access:
    - Last date/time a particular table was accessed
    - Number of times a particular table was accessed in last 3 months

Aggregations and filters (How would the sponsor like to see the data?):

Summary of access by unit

Access by data mart > role

Various ways of differentiating users

Access/usage by org > department

Who shouldn't have access? Retired, or Inactive?

Who is not using their access?

Users/roles who have not accessed a role in 3 months.

Account access by user accounts versus application accounts.

Last access by role

Usage by role

Source Data:

For the purpose of this proof of concept, there are 3 months of raw audit data and security mapping data available in the class database representing three of the core data marts in Cornell's Data Warehouse environment: Finance, Contributor Relations, and Student. There are also tables which contain database account to role mapping, role to table mappings, and a table containing data mart to role mappings. Finally, there is a table containing account names and the associated college/org, and department mappings. Note the account owner names are de-identified (not the actual netids) and the names are AI generated.

You will find the source data in the class database INFO4555 in the *audit_src* schema. If you are unable to view the data, please verify you signed the non-disclosure agreement in Canvas.